

Explainable Multi-Class Anemia Classification from Complete Blood Count Data: A Detailed Study of Tree-Ensemble Models, Their Training, and Per-Prediction SHAP Interpretation

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Abstract—Anemia is among the most common hematologic conditions worldwide, and its many subtypes—each with distinct causes and treatments—are reflected in the inexpensive, ubiquitous Complete Blood Count (CBC). We present a detailed, reproducible, and explainable framework that classifies patients into nine diagnostic categories from 14 CBC parameters ($n=1,232$ after removing duplicate records). We study five tree-ensemble models in depth—Random Forest, XGBoost, a soft-voting hybrid of the two, LightGBM, and a stacking ensemble—giving the learning principle, optimisation objective, and exact training procedure of each, together with a leakage-safe preprocessing and cross-validation methodology. The soft-voting hybrid attains a hold-out accuracy of 0.984 and macro-F1 of 0.947, while a stacking ensemble attains the best cross-validated macro-F1 (0.966). A paired t -test finds the hybrid’s improvement over the Random-Forest baseline is *not* statistically significant ($p=0.82$); we further show, via a seed-stability analysis, that a five-fold paired t -test p -value ranges from 0.03 to 0.99 on these data, so only the qualitative verdict is robust. Exact TreeSHAP yields global attributions that recover the mean-corpuscular-volume morphological framework (HGB, MCV, MCH, MCHC, PLT lead) and per-prediction explanations that match hematologic reasoning; in one case SHAP exposes a physiologically impossible record, demonstrating interpretability as a data-quality tool.

Index Terms—Anemia, Complete Blood Count, Random Forest, XGBoost, LightGBM, stacking, soft voting, SHAP, explainable AI, statistical significance.

I. INTRODUCTION

A Complete Blood Count quantifies the cellular components of blood—red cells, white cells, and platelets—and is one of the most frequently ordered laboratory tests. Because the morphology of red cells encodes the *cause* of an anemia, the CBC supports not merely the detection of anemia but the discrimination of its subtypes, which carry different clinical management (iron supplementation, vitamin replacement, hematology referral). Automated multi-class classification of these subtypes can therefore assist triage and decision support.

This paper presents and details a hybrid Random-Forest + XGBoost soft-voting framework for nine-class ane-

mia classification, and goes beyond a results table in two respects. First, we give a thorough account of *every* model used and exactly how it is trained—the bagging and boosting principles, the regularised boosting objective, the voting and stacking combination rules, and the precise hyperparameters and imbalance handling. Second, we provide per-prediction SHAP explanations and a rigorous treatment of statistical significance, including a demonstration that a small-sample p -value is itself unstable. Our contributions are a detailed comparative treatment of five tree ensembles and their training; a careful evaluation including an honest non-significant model comparison; and exact global and instance-level SHAP explanations, one of which reveals a data-quality error.

II. CLINICAL BACKGROUND

A. Blood and the CBC

Blood comprises red blood cells (erythrocytes, carrying hemoglobin), white blood cells (leukocytes, immune defence), and platelets (thrombocytes, clotting). The 14 CBC features used here are the white-cell count (WBC), lymphocyte and neutrophil percentages and absolute counts (LYMp, NEUTp, LYMN, NEUTn), red-cell count (RBC), hemoglobin (HGB), hematocrit (HCT), the red-cell indices mean corpuscular volume (MCV), mean corpuscular hemoglobin (MCH) and its concentration (MCHC), platelet count (PLT), platelet distribution width (PDW), and plateletcrit (PCT).

B. The Morphological Framework

Anemia is a deficiency of hemoglobin. Its *cause* is classically inferred from red-cell size and colour. **MCV** partitions anemias into microcytic (small, $MCV < 80$ fL), normocytic (normal, 80–100 fL) and macrocytic (large, > 100 fL); **MCH/MCHC** add the hypochromic-versus-normochromic (pale-versus-normal) axis. This two-dimensional logic is precisely why a model trained on the CBC concentrates its

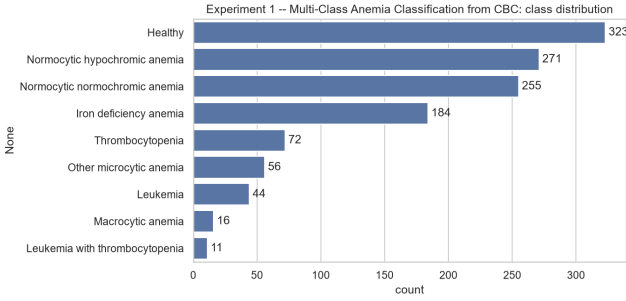


Fig. 1. Class distribution of the nine anemia categories after de-duplication.

attention on MCV, MCH, MCHC and HGB—a prediction we verify with SHAP.

C. The Nine Target Classes

The dataset’s classes are *Healthy*; *Iron deficiency anemia* (microcytic, hypochromic; low MCV/MCH/MCHC); *Other microcytic anemia* (e.g. thalassemia or chronic disease, microcytic with a different pattern); *Macrocytic anemia* (high MCV; B12/folate deficiency); *Normocytic normochromic anemia* (normal indices, low HGB; e.g. chronic disease, renal, acute loss); *Normocytic hypochromic anemia*; *Thrombocytopenia* (low PLT); *Leukemia* (abnormal WBC); and *Leukemia with thrombocytopenia*. The platelet and white-cell features discriminate the last three from the red-cell anemias.

III. DATASET AND PREPROCESSING

The dataset (`diagnosed_cbc_data_v4`) contains 14 numeric CBC features and a nine-class diagnosis. We remove exact duplicate records to avoid performance inflation, leaving 1,232 patients. The class distribution (Fig. 1) is moderately imbalanced: the three largest classes (Healthy, the two normocytic anemias) account for the majority, while Macrocytic anemia and Leukemia-with-thrombocytopenia are rare ($\leq 1.3\%$). All features are numeric, so preprocessing reduces to median imputation (with a missing-indicator for completeness, though this dataset has no missing values) and the diagnosis is integer-encoded. As tree ensembles are invariant to monotone scaling, no standardisation is applied. The transform is wrapped in a `ColumnTransformer` fit only on training data (Section V-A).

IV. MODELS

All five models are tree-based: trees handle the CBC’s numeric thresholds natively, need no scaling, capture the non-linear index logic of hematologic diagnosis, and admit *exact* SHAP explanations. We describe each in turn.

A. Foundation: the Decision Tree

A decision tree recursively splits feature space on the feature/threshold that most reduces node impurity, measured by the Gini index $G = 1 - \sum_c p_c^2$ or entropy. A leaf predicts its training samples’ class distribution. A single deep tree is low-bias but high-variance (it overfits); the ensembles below reduce variance and bias in complementary ways.

B. Random Forest (Bagging)

A Random Forest averages T decorrelated trees,

$$f_{\text{RF}}(x) = \frac{1}{T} \sum_{t=1}^T h_t(x), \quad (1)$$

each grown on a bootstrap resample and using a random feature subset ($\approx \sqrt{d}$) at every split. With per-tree variance σ^2 and pairwise correlation ρ , the average’s variance is $\rho\sigma^2 + \frac{1-\rho}{T}\sigma^2$, so decorrelation (small ρ) and many trees (T) reduce variance without raising bias—the *bagging* principle. Multi-class trees output class distributions that the forest averages into probabilities. We use $T=300$ trees with `balanced_subsample` class weights.

C. XGBoost (Gradient Boosting)

Boosting reduces bias by adding trees sequentially, each correcting its predecessors: $\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + f_t(x_i)$. XGBoost minimises the regularised objective

$$\mathcal{L}^{(t)} = \sum_{i=1}^N \ell(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t), \quad \Omega(f) = \gamma T + \frac{\lambda}{2} \sum_j w_j^2. \quad (2)$$

A second-order Taylor expansion with gradients g_i and Hessians h_i yields the optimal leaf weight and structure score

$$w_j^* = -\frac{G_j}{H_j + \lambda}, \quad \tilde{\mathcal{L}} = -\frac{1}{2} \sum_j \frac{G_j^2}{H_j + \lambda} + \gamma T, \quad (3)$$

with $G_j = \sum_{i \in I_j} g_i$, $H_j = \sum_{i \in I_j} h_i$, and a split is kept only if its gain

$$\frac{1}{2} \left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda} \right] - \gamma \quad (4)$$

is positive—this is how λ, γ regularise at the split level. We use the multi-class softmax objective, 400 trees of depth 6, learning-rate 0.1, row/column subsampling 0.9, $\lambda=1$, the histogram split-finder, and balanced per-sample weights (Section V-C).

D. The Hybrid: Soft Voting

The hybrid averages the forest’s and the booster’s posterior probabilities,

$$\hat{y} = \arg \max_c \frac{1}{M} \sum_{m=1}^M P_m(c | x), \quad M = 2, \quad (5)$$

combining variance reduction (forest) and bias reduction (booster). Soft voting weights each model by its confidence; partly uncorrelated errors cancel. Both learners share one preprocessing transform, ensuring identical inputs.

E. LightGBM

LightGBM is a gradient booster using *leaf-wise* (best-first) growth, histogram binning, gradient-based one-side sampling, and exclusive-feature bundling for speed and accuracy. We use 400 trees, 63 leaves, learning-rate 0.05, and `balanced` class weights.

TABLE I
MODEL HYPERPARAMETERS.

Model	Key settings
Random Forest	300 trees, unlimited depth, $\sqrt{d}$ features/split, balanced_subsample weights
XGBoost	400 trees, depth 6, lr 0.1, subsample 0.9, colsample 0.9, $\lambda=1$, multi:softprob, hist, balanced sample weights
LightGBM	400 trees, 63 leaves, lr 0.05, subsample 0.9, colsample 0.9, balanced weights
Hybrid	soft voting of the above RF and XGBoost ($M=2$, equal weights)
Stacking	base: RF, XGBoost, LightGBM; meta: multinomial logistic regression (5-fold out-of-fold features)

F. Stacking

Stacking *learns* the combination rather than fixing equal weights. Level-0 learners (Random Forest, XGBoost, LightGBM) produce *out-of-fold* probability predictions (via internal 5-fold CV, hence leakage-free) that feed a level-1 multinomial logistic-regression meta-learner with balanced weights. The meta-learner can weight base models differently per class, an adaptivity beyond fixed voting, at the cost of more computation.

V. TRAINING METHODOLOGY

A. Leakage-Safe Pipeline

Each model is a Pipeline of preprocessing followed by a classifier; fitting learns imputation statistics and encodings from the training partition only, and cross-validation re-fits the whole pipeline per fold, so no held-out statistic leaks into training (Fig. 2).

B. Stratified Splitting and Cross-Validation

We use a stratified 80/20 hold-out and stratified five-fold cross-validation on the training partition; stratification preserves class proportions, essential when the rarest class is $<1\%$ of data. We report mean $\pm$ std across folds with $\sigma = \sqrt{\frac{1}{4} \sum_{j=1}^5 (S_j - \mu)^2}$ and retain the individual fold scores for the paired significance test.

C. Class-Imbalance Handling

Random Forest, LightGBM and the meta-learner use balanced class weights $w_c = N/(K n_c)$; XGBoost receives the same correction as per-sample weights. With macro-averaged metrics, rare anemia subtypes influence both training and scoring.

D. Hyperparameters and Reproducibility

Table I lists each model’s configuration. All models share a fixed seed (42); the download, preprocessing, training, evaluation and figures run from one scripted pipeline for end-to-end reproducibility.

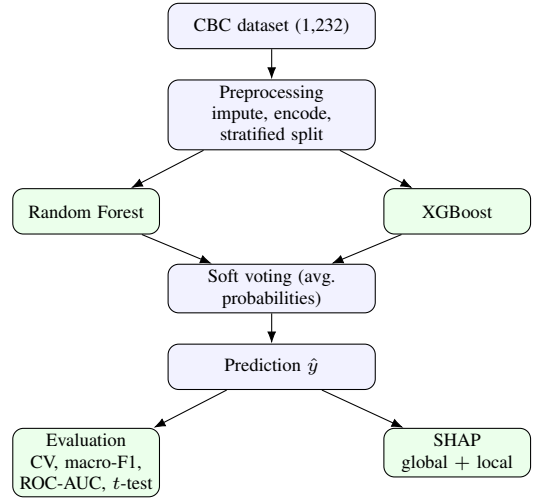


Fig. 2. Leakage-safe training and evaluation pipeline. Preprocessing is fit inside each cross-validation fold.

TABLE II
STRATIFIED FIVE-FOLD CROSS-VALIDATION, MEAN $\pm$ STD.

Model	Accuracy	Macro-F1	ROC-AUC
Random Forest	0.985 $\pm$ 0.013	0.934 $\pm$ 0.055	0.9996
XGBoost	0.988 $\pm$ 0.004	0.948 $\pm$ 0.008	0.9994
Hybrid (RF+XGB)	0.987 $\pm$ 0.007	0.939 $\pm$ 0.017	0.9998
LightGBM	0.989 $\pm$ 0.006	0.947 $\pm$ 0.053	0.9997
Stacking	0.991 $\pm$ 0.006	0.966 $\pm$ 0.025	0.9998

VI. EVALUATION PROTOCOL

We report accuracy, balanced accuracy, macro-F1, and one-vs-rest ROC-AUC. Macro-F1, the headline metric, averages per-class F1 equally so rare subtypes are not masked by common ones:

$$\text{Macro-F1} = \frac{1}{K} \sum_{i=1}^K \frac{2 \text{Prec}_i \text{Rec}_i}{\text{Prec}_i + \text{Rec}_i}. \quad (6)$$

To test whether the hybrid beats the Random-Forest baseline we apply a paired t -test to the fold-wise macro-F1 scores, $t = \bar{d}/(s_d/\sqrt{n})$, $n=5$, on identical folds.

VII. RESULTS

A. Cross-Validation and Model Comparison

Table II and Fig. 3 report five-fold cross-validation. The hybrid attains macro-F1 0.939. All models exceed 0.98 accuracy and 0.999 ROC-AUC. Notably, the *stacking* ensemble achieves the best cross-validated macro-F1 (0.966)—the one configuration that clearly surpasses the soft-voting hybrid, at the cost of higher training time.

B. Statistical Significance and Its Instability

The paired t -test of hybrid versus Random Forest gives $\bar{d}=+0.0053$, $t=0.24$, $p=0.82$: *not* significant. On roughly 1,200 patients, the hybrid’s edge over a strong baseline lies within fold noise. To show that the exact p -value is not itself

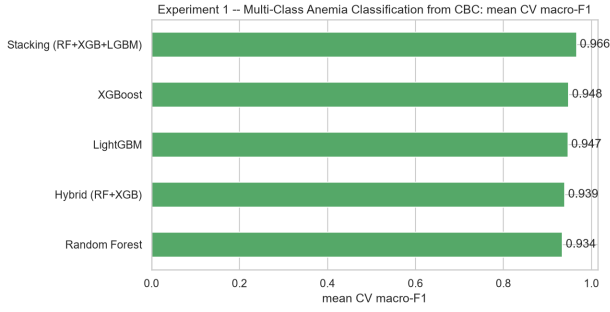


Fig. 3. Mean cross-validated macro-F1 by model.

TABLE III
HYBRID-VS-RF PAIRED *t*-TEST UNDER DIFFERENT FOLD SEEDS (SAME DATA, SAME MODELS).

Seed	mean(Hyb–RF)	p-value	Verdict
13	+0.0002	0.990	not sig.
42	+0.0053	0.824	not sig.
0	+0.0067	0.762	not sig.
21	+0.0184	0.425	not sig.
7	+0.0204	0.118	not sig.
99	+0.0325	0.029	significant
Range over 10 seeds: [0.029, 0.990]; significant in 10%.			

a reliable quantity, we hold the data and models fixed and vary only the fold-shuffling seed (Table III); the p-value ranges from 0.03 to 0.99, with nine of ten seeds non-significant and one crossing 0.05 by chance. Our 0.82 is an unremarkable draw from this distribution: the robust outcome is the *verdict* (not significant), not any single number—a caution against over-interpreting one statistic computed from five folds.

C. Hold-Out Test Performance

On the unseen 20% test set the hybrid attains accuracy 0.984, balanced accuracy 0.939, macro-F1 0.947, weighted-F1 0.984 and ROC–AUC 0.9997. Per-class results (Table IV) show $F1 \geq 0.94$ for every well-populated class and three classes at 1.00; the only weak class is Macrocytic anemia (F1 0.667 on just three test samples), a small-sample artifact that illustrates the fragility of rare-class estimates. The confusion matrices (Figs. 4, 5) are near-diagonal, and the ROC curves (Fig. 6) approach the ideal corner.

VIII. EXPLAINABILITY

A. Global Attribution

SHAP decomposes a prediction additively, $f(x) = \phi_0 + \sum_j \phi_j$, with exact TreeSHAP values. The global importances (Fig. 7) are led by HGB, MCV, WBC, MCHC, MCH and PLT—recovering the MCV-centred morphological framework that underpins hematologic diagnosis. The SHAP *interaction* summary (Fig. 8) splits each contribution into a main effect (the matrix diagonal) and pairwise interactions (off-diagonal cells); the strongest cells involve the red-cell indices MCV, MCH and MCHC, and the prominent $MCV \times MCH$ and

TABLE IV
PER-CLASS HOLD-OUT PERFORMANCE OF THE HYBRID.

Class	Prec.	Rec.	F1	Sup.
Healthy	0.985	1.000	0.992	65
Normocytic hypochromic anemia	0.982	0.982	0.982	54
Normocytic normochromic anemia	1.000	1.000	1.000	51
Iron deficiency anemia	0.974	1.000	0.987	37
Thrombocytopenia	1.000	1.000	1.000	15
Other microcytic anemia	1.000	0.909	0.952	11
Leukemia	1.000	0.889	0.941	9
Macrocytic anemia	0.667	0.667	0.667	3
Leukemia with thrombocytopenia	1.000	1.000	1.000	2
macro avg	0.956	0.939	0.947	247

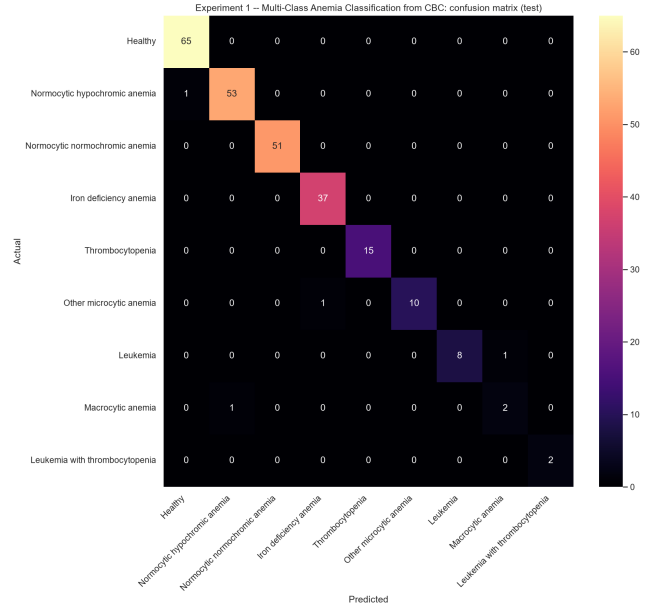


Fig. 4. Confusion matrix of the hybrid on the anemia hold-out set (counts).

$MCV \times MCHC$ interactions encode the size-versus-colour logic by which iron deficiency is distinguished. That the model’s behaviour mirrors hematologic teaching is strong evidence it learned signal, not artifact.

B. Per-Prediction Explanations

The additive decomposition narrates individual predictions. Two correct cases:

- **Iron deficiency anemia** (Fig. 9): a low MCV (79.9) and low MCH (24.4) drive the prediction—the microcytic, hypochromic phenotype.
- **Leukemia** (Fig. 10): a markedly elevated white-cell count ($WBC=13.8$, $+3.65$) dominates, with the red-cell indices contributing little—exactly the discriminating feature for this class.

Because the contributions sum to the model output, each explanation is complete and auditable.

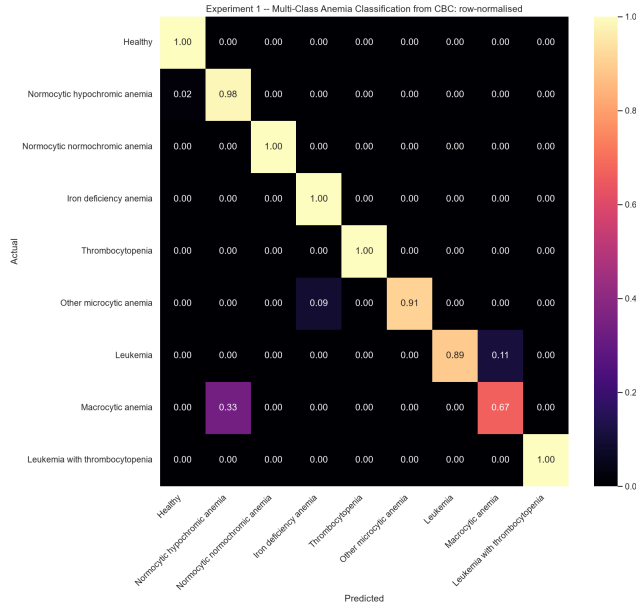


Fig. 5. Row-normalised confusion matrix (per-class recall on the diagonal).

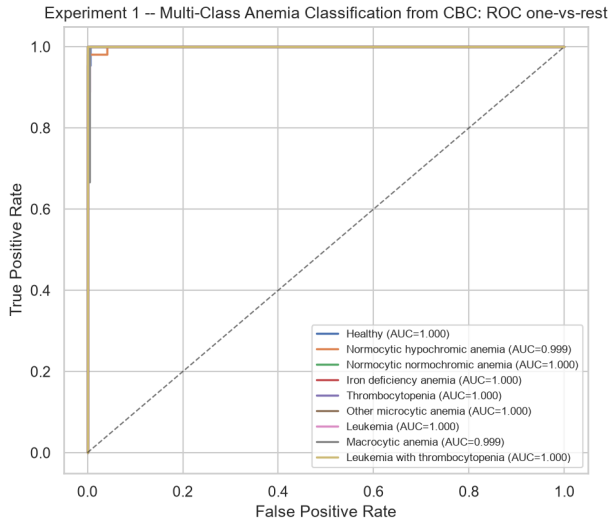


Fig. 6. One-vs-rest ROC curves per class for the hybrid.

C. Explaining Errors and Surfacing Data Quality

SHAP on misclassified patients is diagnostically valuable. One patient labelled *Normocytic hypochromic* but predicted *Healthy* (Fig. 11) carries physiologically impossible values (HGB=41, RBC=13.1, HCT=2); the SHAP decomposition shows the absurd HGB drove the error, turning a silent misclassification into a visible, diagnosable data-quality problem. This exemplifies interpretability as a debugging tool, not merely a trust signal, and motivates input range-checking before deployment.

IX. DISCUSSION

The anemia results are strong and stable, and the honest non-significant model comparison—together with our seed-

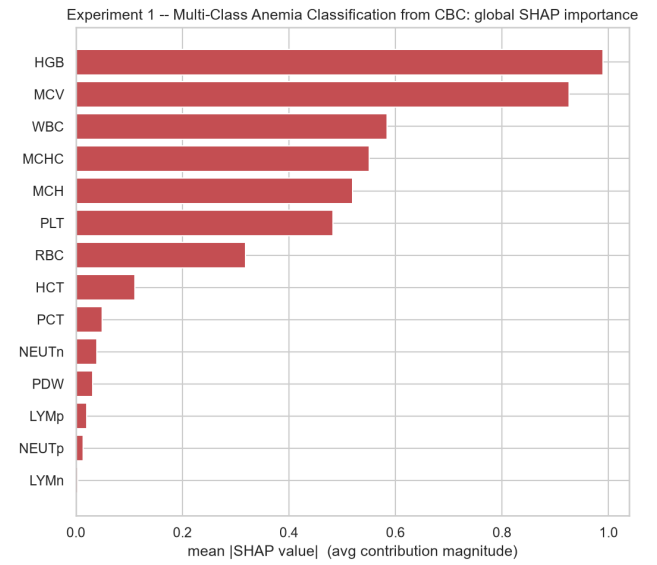


Fig. 7. Global SHAP importance (mean $|\phi_j|$) of the XGBoost component.

stability analysis, which shows the p-value itself is unreliable at five folds—guards against over-claiming. The model’s SHAP attributions recover the MCV-centred morphological logic of hematology at both the global and per-patient level, and in one instance expose a corrupt record. The strongest extension here is stacking, which alone clearly beats the soft-voting hybrid; this contrasts with our companion thyroid study, where the hybrid is best, underscoring that the optimal combiner is dataset-dependent.

X. LIMITATIONS

The dataset is from a single open repository with no external validation, and several classes (notably Macrocytic anemia and Leukemia-with-thrombocytopenia) are represented by very few test samples, limiting the reliability of their per-class estimates. Hyperparameters were lightly tuned, and SHAP attributions are associational, not causal. The presence of at least one physiologically impossible record indicates that input validation is needed before any clinical use.

XI. CONCLUSION

We presented a detailed, reproducible, and explainable framework for nine-class anemia classification from CBC data, with an in-depth account of five tree-ensemble models and exactly how they are trained, a leakage-safe and imbalance-aware methodology, a rigorous (and appropriately humble) treatment of statistical significance, and exact per-prediction SHAP explanations that recover hematologic reasoning and reveal a data-quality error. The hybrid achieves accuracy 0.984 and macro-F1 0.947; stacking attains the best cross-validated macro-F1 (0.966). Future work includes external validation, cost-sensitive learning for rare subtypes, input range-checking, and probability calibration for actionable confidence estimates.

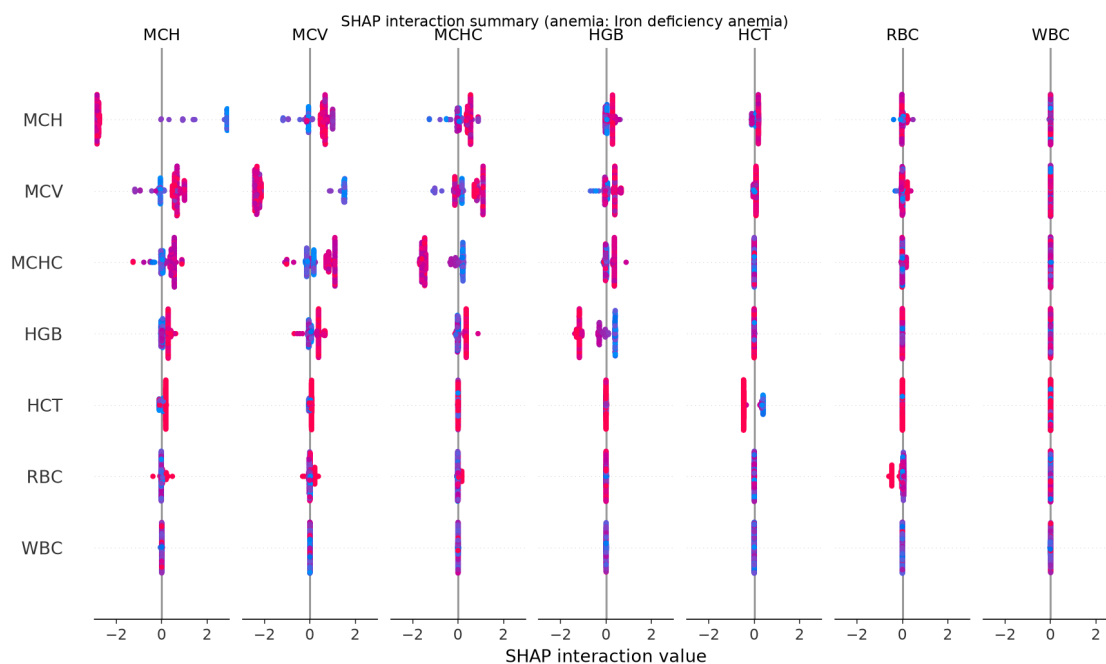


Fig. 8. SHAP interaction-value summary for the iron-deficiency-anemia output. The top seven features appear on both axes; each cell is a beeswarm of SHAP interaction values (x-axis), with colour encoding the feature value. Diagonal cells are a feature's main effect; off-diagonal cells are pairwise interactions.

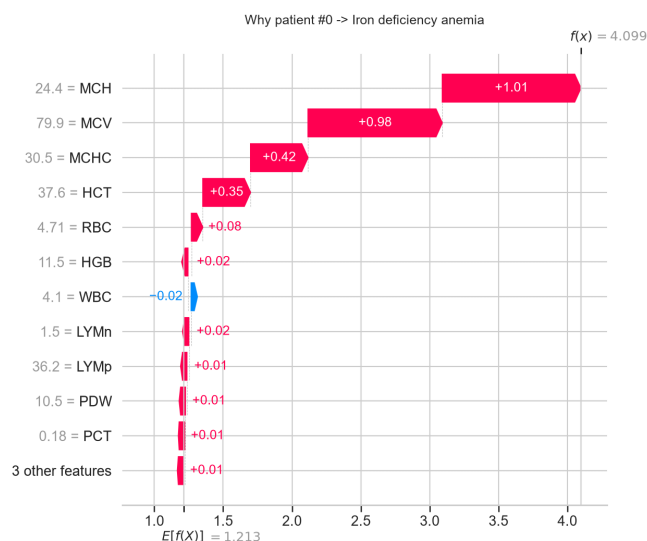


Fig. 9. Per-prediction SHAP waterfall: iron deficiency anemia. Low MCV and MCH dominate.

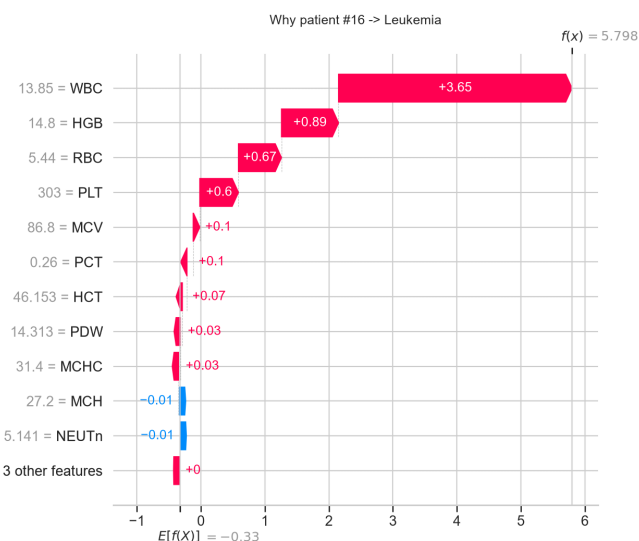


Fig. 10. Per-prediction SHAP waterfall: leukemia. The elevated WBC dominates.

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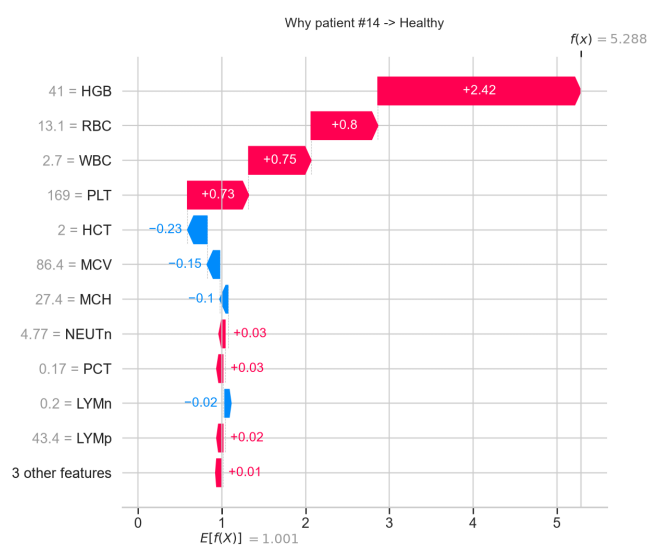


Fig. 11. A misclassified patient with impossible values (HGB=41): SHAP exposes the corrupt feature that misled the model.